PrivaGene

Private Set Intersection for Healthcare Application
Group FYP-25-S4-17

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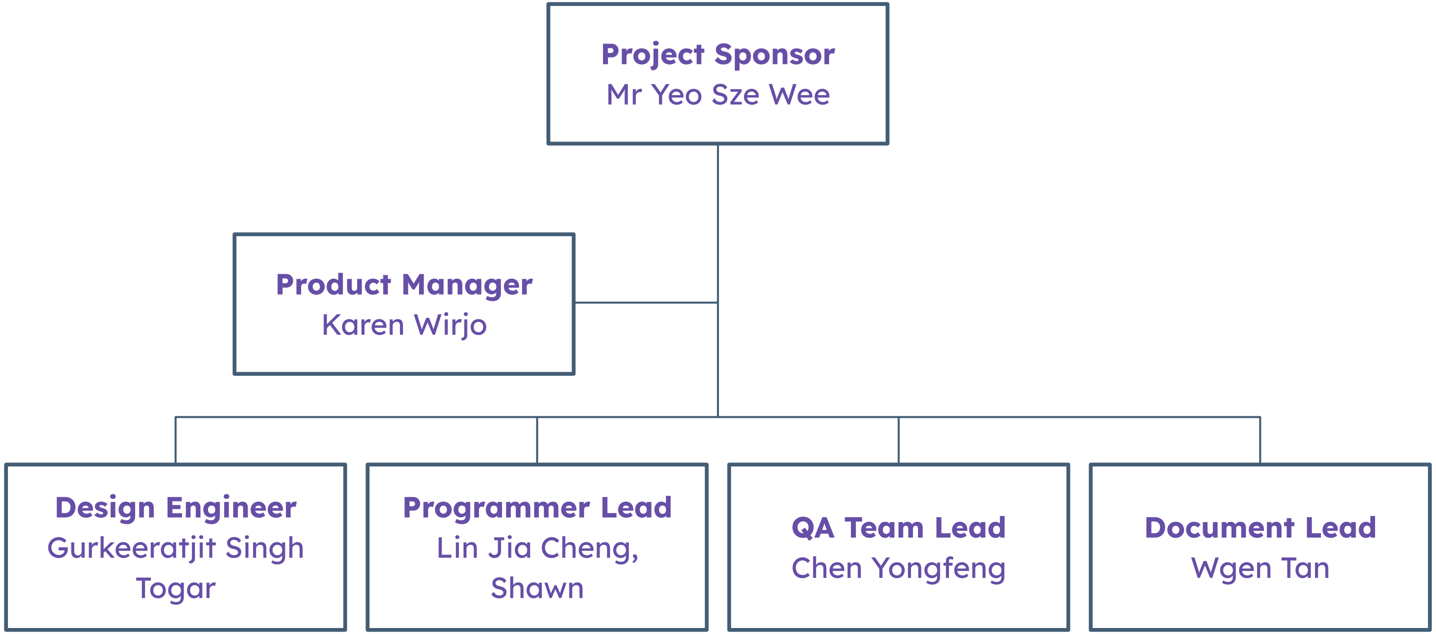
Introduction To The Team



01

Recap & What Changed

Quick Recap & Progress

What is PrivaGene?

A privacy-preserving genetic risk assessment platform using Diffie-Hellman Private Set Intersection (PSI). Patients assess disease risk without exposing raw genetic data to hospitals, while hospitals provide assessments without revealing their proprietary disease-gene databases.

What Changed Since Last Time?

Since the prototype presentation, we have: enhanced caregiver delegated assessments, improved UI/UX polish across all 7 user roles, pursued real-world clinical data from 5 Singapore medical institutions, deployed the full system to production on Render, and conducted comprehensive system testing.

New & Enhanced Features

Caregiver Assessments

Caregivers can now initiate risk assessments on behalf of patients with explicit authorization, enabling family-managed care workflows

Production Deployment

Full system deployed to Render cloud platform with frontend and backend services running in production environment

Data Acquisition

Contacted 5 Singapore medical institutions for clinical genetic datasets and investigated ClinVar/gnomAD public databases

Risk Results Exporting

Patients who want to share their risk results with a professional can now export their PSI risk assessment results.

Security Hardening

We added AES-256 encryption at rest for the risk analysis, as well as role-based access middleware, comprehensive audit logging to ensure added privacy.

UI/UX Polish

Refined interfaces across all pages spanning 7 user roles with consistent design language and improved navigation. Added search and filter functions to the relevant pages.

02

Data & Deployment

Real-World Data Acquisition

Institutions Contacted

We contacted 5 Singapore medical/clinical institutions: Singapore General Hospital (SGH), A*STAR Genome Institute of Singapore (GIS), CRIS/PRECISE, NUHS, and Mount Elizabeth Hospital. Only CRIS responded.

They confirmed that accessing datasets such as PRECISE-SG100K requires a formal Call for Proposals with a senior researcher as Lead PI, which is beyond the scope of our capabilities as students. They did however point us to publicly available databases.

Alternative Datasets

Following CRIS's recommendation, we investigated gnomAD and ClinVar. gnomAD provides aggregate population-level variant data, not individual-level datasets. ClinVar offers gene-disease association mappings useful for enriching our disease database.

However, neither replaces real patient data so we validated using controlled test data with known expected outcomes instead.

Deployment: Localhost to Production

Frontend

Vanilla HTML/CSS/JS frontend deployed to Render static site. Dynamic API base URL configuration via config.js for seamless local/production switching.

Backend

Node.js/Express backend deployed as a Render web service. CORS middleware configured for cross-origin requests between frontend and backend domains.

Database

SQLite database deployed on Render with persistent disk. Migrations run automatically on server startup for seamless schema management.

Challenges

Free-tier cold starts (30-60s spin-up), database resets on redeployment, environment variable management across local and production configs.

03

Testing & Validation

System Test Results Summary

Test Category	Test Cases	Passed	Status	Key Tests
PSI Computation	7	7	All Pass	• Gene blinding correctness • Modular exponentiation accuracy • Risk percentage calculation
Authentication & RBAC	16	16	All Pass	• Role-based access controls • Login/register validation • Session enforcement
Data & Security	20	20	All Pass	• AES-256 encryption at rest • Audit log integrity • Bcrypt password hashing
Functional Modules	97	97	All Pass	• Research analytics • Caregiver permissions
Unit Tests (Backend Services)	38	38	All Pass	• Encryption service • PSI service • Middleware validation

PSI Correctness Validation

Mathematical Verification

Verified that:

H(gene)^(ab) mod P = H(gene)^(ba) mod P

for all test inputs. Confirmed intersection correctness: only matching genes between patient set X and disease set Y are identified. Risk formula $c|X \cap Y|/n$ % produces expected percentages across all test scenarios.

Edge Case Testing

Tested with:

- empty patient gene set (0% risk),
- no matching genes (0% risk),
- all genes matching (maximum risk),
- single gene match,
- and large gene sets.

Each run generates fresh random secrets, ensuring different blinded values but identical final intersection results.

04

Demo & Future Roadmap

Future Roadmap

Short-Term

- Integrate ClinVar gene-condition mappings for clinically-grounded disease data.
- Implement PDF export for risk results.
- Add multi-factor authentication for hospital administrators.

Medium-Term

- Blockchain-based audit log anchoring for tamper-proof compliance records.
- Research consent tracking on-chain. Notification system for new assessment results.

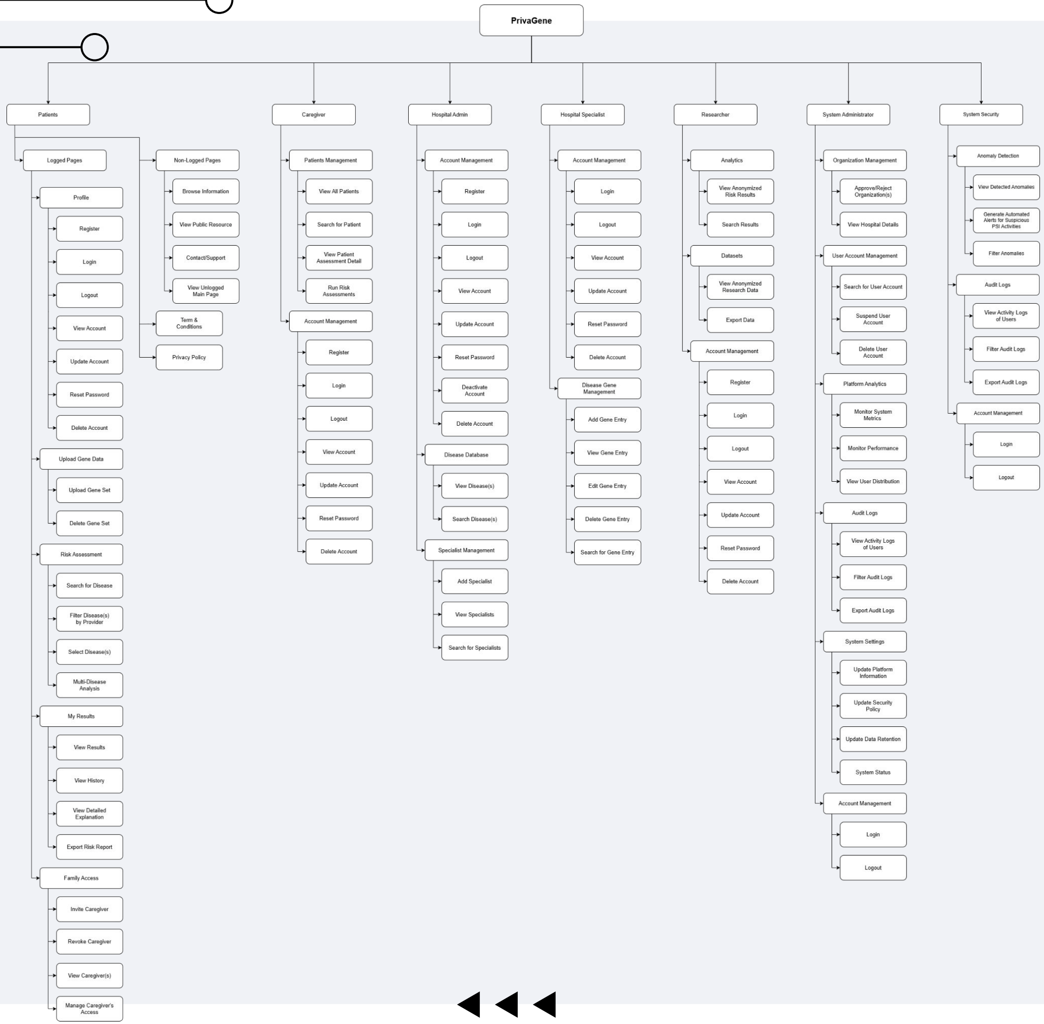
Long-Term

- Encrypted data server separation. Hospital credential verification via blockchain.
- Appointment booking integration. Scalable cloud infrastructure beyond free tier.

put

WorkBreakdown Structure

here



PrivaGene
Patients
Logged Pages
Profile
Register
Login
Logout
View Account
Update Account
Reset Password
Delete Account
Upload Gene Data
Upload Gene Set
Delete Gene Set
Risk Assessment
Search for Disease
Filter Disease(s) by Provider
Select Disease(s)
Multi-Disease Analysis
My Results
View Results
View History
View Detailed Explanation
Export Risk Report
Family Access
Invite Caregiver
Revoke Caregiver
View Caregiver(s)
Manage Caregiver's Access
Non-Logged Pages
Browse Information
View Public Resource
Contact/Support
View Unlogged Main Page
Term & Conditions
Privacy Policy
Caregiver
Patients Management
View All Patients
Search for Patient
View Patient Assessment Detail
Run Risk Assessments
Account Management
Register
Login
Logout
View Account
Update Account
Reset Password
Delete Account
Hospital Admin
Account Management
Register
Login
Logout
View Account
Update Account
Reset Password
Deactivate Account
Delete Account
Disease Database
View Disease(s)
Search Disease(s)
Specialist Management
Add Specialist
View Specialists
Search for Specialists
Hospital Specialist
Account Management
Login
Logout
View Account
Update Account
Reset Password
Delete Account
Disease Gene Management
Add Gene Entry
View Gene Entry
Edit Gene Entry
Delete Gene Entry
Search for Gene Entry
Researcher
Analytics
View Anonymized Risk Results
Search Results
Datasets
View Anonymized Research Data
Export Data
Account Management
Register
Login
Logout
View Account
Update Account
Reset Password
Delete Account
System Administrator
Organization Management
Approve/Reject Organization(s)
View Hospital Details
User Account Management
Search for User Account
Suspend User Account
Delete User Account
Platform Analytics
Monitor System Metrics
Monitor Performance
View User Distribution
Audit Logs
View Activity Logs of Users
Filter Audit Logs
Export Audit Logs
System Settings
Update Platform Information
Update Security Policy
Update Data Retention
System Status
Account Management
Login
Logout
System Security
Anomaly Detection
View Detected Anomalies
Generate Automated Alerts for Suspicious PSI Activities
Filter Anomalies
Audit Logs
View Activity Logs of Users
Filter Audit Logs
Export Audit Logs
Account Management
Login
Logout

Demo

https://privagene.onrender.com/pages/index.html

51

Frontend pages across 7 user roles

80+

Backend API endpoints

45+

System test cases passed

Project Timeline

Oct: Project kickoff & requirements

Nov: Core PSI implementation

Dec: Prototype presentation

Dec: Production deployment

Jan: Data acquisition outreach

Jan: Testing & validation

Feb: Final polishing

Feb 14: Final submission

Thank You